Orders Interface Specifications

Product Name: HistoTrac

Component Name: SystemLink Orders Service

Version: 2.0

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1. Introduction

This specification details the structure for an inbounds order interface. This interface allows a third-party system to send an order to HistoTrac that will generate an order in HistoTrac. The orders are recorded in HistoTrac and are placed in a queue for review before they are released into the HistoTrac system for accessioning/resulting.

1.1 Business Case

Third party systems that receive or generate orders that could be passed to a laboratory utilizing HistoTrac are currently manually entered into HistoTrac. Since these orders have already been placed in a third-party system it is more efficient to pass the order to HistoTrac through an automated interface. By utilizing the interface, the orders will no longer need to be entered into HistoTrac reducing transcription errors and delays in getting paperwork to the laboratory. Orders received through the interface in HistoTrac will be placed in a queue for review. Once the reviews are accepted the order will enter the workflow within the HistoTrac database.

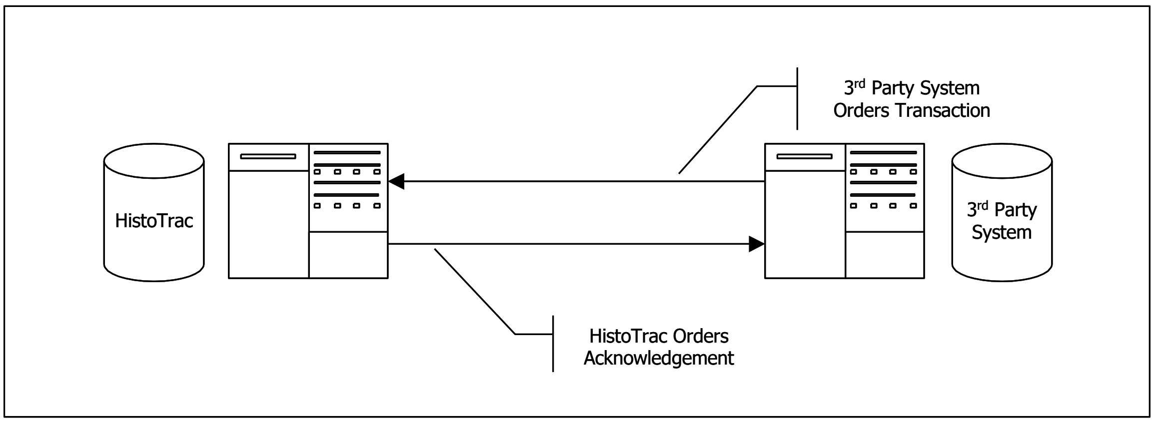
1.2 Software Change Intended Use

The Orders Interface is being developed so that third party software systems can send orders directly to the HistoTrac repository.

2. General Workflow

With the Orders interface the third-party system will send a correctly formatted HL7 orders transaction to a specific IP Address and port. The HistoTrac HL7 Manager will be listening for orders on the specified IP address and port. When a message is received an acknowledge message is created and is transmitted back to the sending IP Address and port. This acknowledgement transaction signals the remote system that HistoTrac is ready to accept a new order transaction.

Data Flow



3. HistoTrac and 3rd Party System Communication

3.1 Communication Protocol and Transaction Framing

All transactions will be sent using TCP/IP as the communication protocol. With the reliability of TCP, no block check calculations or data length counts are necessary. The following applies to the communication envelope for all messages. The characters listed in <> below represent ASCII characters.

- All messages and acknowledgments will begin with <11>
- All segments will end with <13>
- All messages and acknowledgments will terminate with <28><13>

3.2 Acknowledgments (ACK/NAK)

The receiving system (HistoTrac/server) must return a communication level acknowledgment to the sending system (3rd party system/client) for every message that is received. Once the client has received the acknowledgment, it may then send the next message. For example, when an orders transaction is sent to HistoTrac, HistoTrac will return an acknowledgment to the originating system for that message on the same logical socket where it was received. The other system can then send the next order transaction to HistoTrac.

Acknowledgments can be either positive (ACK) or negative (NAK). For all ACK/NAK messages, the HL7 segments MSH and MSA must be returned. The MSA; 1 is valued with "CA" for ACK and with "CE" for NAK.

4. HistoTrac HL7 Order Transaction and Segments

4.1 ORM – General Order Message

The Orders transaction (ORM – General Order Message) is separated into the following five segments:

MSH – Message Header
PID – Patient Identification
PV1 – Patient Visit
ORC – Common Order
[{NTE}] – Notes and Comments

The following table outlines the contents of each segment. Each segment is required in the order transaction.

Legend
R – Required
O – Optional
NS – Not Supported

4.1.1 MSH – Message Header

Sequence #	Name	Max Length	Field Requirement	Notes
1	Field Separator	1	R	|
2	Encoding Characters	4	R	component delimiter (^) repeat delimiter (~) escape character (\) subcomponent delimiter (&)
3	Sending Application	50	R	
4	Sending Facility	50	O	
5	Receiving Applications	50	O	
6	Receiving Facility	50	O	
7	Date/Time of Message	12	R	CCYYMMDDHHMM
8	Security	N/A	NS	
9	Message Type	7	R	Contains message type and trigger event.
10	Message Control ID	15	R	This sequence is valued with a number that is unique to that message. The receiving system will echo this value in the MSA message.
11	Processing ID	3	R	T for Training or P for Production
12	Version ID	8	R	Specifies the HL7 version in use
13	Sequence Number	N/A	NS	
14	Continuation Pointer	N/A	NS	
15	Accept Acknowledgement Type	N/A	NS	

16	Application Acknowledgement Type	N/A	NS	
17	Country Code	N/A	NS	
18	Character Set	N/A	NS	
19	Principal Language of Message	N/A	NS	

4.1.2 PID – Patient Identification

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID – Patient ID	N/A	NS	
2	Patient ID (External ID)	N/A	NS	
3	Patient ID (Internal ID)	20	R	Component 1 = Medical Record Number Component 2 = "” Component 3 = MulHos Code Example: Z442345^^FZQ
4	Alternate Patient ID	20	O	
5	Patient's Name	55	R	Component 1=LAST NAME Component 2=FIRST NAME Component 3=MIDDLE INITIAL Example: SMITH^JOHN^D
6	Mother's Maiden Name	40	O	
7	Date of Birth	8	O	CCYYMMDD
8	Sex	15	O	
9	Patient Alias	N/A	NS	
10	Race	15	O	
11	Patient Address	40, 40, 30, 2, 10	O	Component 1=address line 1 Component 2=address line 2 Component 3=city Component 4=state Component 5=zip code Example: 123 Oak Street ^ Suite 100 ^ Anywhere ^ VA ^ 12345
12	Country Code	15	O	
13	Phone Number – Home	15	O	
14	Phone Number – Business	15	O	
15	Language – Patient	N/A	NS	
16	Marital Status	N/A	NS	
17	Religion	N/A	NS	
18	Patient Account Number	25	O	
19	SSN Number – Patient	15	O	
20	Driver's License – Patient	N/A	NS	
21	Mother's Identifier	N/A	NS	
22	Ethnic Group	15	O	

23	Birth Place	N/A	NS	
24	Multiple Birth Indicator	N/A	NS	
25	Birth Order	N/A	NS	
26	Citizenship	N/A	NS	
27	Veteran's Military Status	N/A	NS	
28	Nationality	N/A	NS	
29	Patient Death Date/Time	N/A	NS	
30	Patient Death Indicator	N/A	NS	

4.1.3 PV1 – Patient Visit

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID PV1	N/A	NS	
2	Patient Class	1	R	
3	Assigned Patient Location	20, 20, 20	O	Component 1 = Unit Component 2 = Room Component 3 = Bed Example: unit^room^bed
4	Admission Type	N/A	NS	
5	Pre-Admit Number	N/A	NS	
6	Prior Patient Location	N/A	NS	
7	Attending Doctor	10, 25, 25	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
8	Referring Doctor	N/A	NS	
9	Consulting Doctor	N/A	NS	
10	Hospital Service	N/A	NS	
11	Temporary Location	N/A	NS	
12	Pre-Admit Test Indicator	N/A	NS	
13	Re-Admission Indicator	N/A	NS	
14	Admit Source	N/A	NS	
15	Ambulatory Status	N/A	NS	
16	VIP Indicators	N/A	NS	
17	Admitting Doctor	10, 25, 25	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
18	Patient Type	N/A	NS	
19	Visit Number	15, 15	R	Component 1 = Episode ID Component 2 = Event ID Example: 1^2
20	Financial Class	N/A	NS	
21	Charge Price Indicator	N/A	NS	

22	Courtesy Code	N/A	NS	
23	Credit Rating	N/A	NS	
24	Contract Code	N/A	NS	
25	Contract Effective Date	N/A	NS	
26	Contract Amount	N/A	NS	
27	Contract Period	N/A	NS	
28	Interest Code	N/A	NS	
29	Transfer to Bad Debt Code	N/A	NS	
30	Transfer to Bad Debt Date	N/A	NS	
31	Bad Debt Agency Code	N/A	NS	
32	Bad Debt Transfer Amount	N/A	NS	
33	Bad Debt Recovery Amount	N/A	NS	
34	Delete Account Indicator	N/A	NS	
35	Delete Account Date	N/A	NS	
36	Discharge Disposition	N/A	NS	
37	Discharged to Location	N/A	NS	
38	Diet Type	N/A	NS	
39	Servicing Facility	N/A	NS	
40	Bed Status	N/A	NS	
41	Account Status	N/A	NS	
42	Pending Location	N/A	NS	
43	Prior Temporary Location	N/A	NS	
44	Admit Date/Time	8	R	CCYYMMDD
45	Discharge Date/Time	8	O	CCYYMMDD
46	Current Patient Balance	N/A	NS	
47	Total Charges	N/A	NS	
48	Total Adjustments	N/A	NS	
49	Total Payments	N/A	NS	
50	Alternate Visit ID	N/A	NS	
51	Visit Indicator	N/A	NS	
52	Other Health Care Provider	N/A	NS	

4.1.4 ORC – Common Order

Sequence #	Name	Max Length	Field Requirement	Notes
1	Order Control	2	R	
2	Placer Order Number	15, 50	R	Component 1 = Test Code Component 2 = Description Example: ABC^SEROLOGY CLASS I
3	Filler Order Number	15 8	O	Component 1 = Accession Number Component 2= Draw Date (CCYYMMDD) Example: 99H7242^20020802

4	Placer Order Number	N/A	NS	
5	Order Status	15	O	Order Priority
6	Response Flag	N/A	NS	
7	Quantity/Timing	N/A	NS	
8	Parent	N/A	NS	
9	Date/Time of Transaction	8	R	CCYYMMDD
10	Entered By	N/A	NS	
11	Verified By	N/A	NS	
12	Ordering Provider	10, 25, 25	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
13	Enterer's Location	N/A	NS	
14	Call Back Phone Number	N/A	NS	
15	Order Effective Date/Time	N/A	NS	
16	Order Control Code Reason	N/A	NS	
17	Entering Organization	N/A	NS	
18	Entering Device	N/A	NS	
19	Action By	N/A	NS	

4.1.5 OBR – Observation Request

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID	2	O	
2	Placer Order Number	15, 50	R	Component 1 = Test Code Component 2 = Description Example: ABC^SEROLOGY CLASS I
3	Filler Order Number	15 8	O	Component 1 = Accession Number Component 2= Draw Date (CCYYMMDD) Example: 99H7242^20020802
4	Universal Service Identifier	200	R	
5	Priority	15	O	Order Priority
6	Requested Date/Time	N/A	O	CCYYMMDD (optional - time supported)
7	Observation Date/Time	N/A	O	CCYYMMDD (optional - time supported)
8	Observation End Date/Time	N/A	NS	
9	N/A	N/A	N/A	
10	N/A	N/A	N/A	
11	Action Code	N/A	NS	
12	N/A	N/A	NS	
13	Clinical Info	N/A	O	
14	Specimen Received	N/A	O	CCYYMMDDHHMMSS
15	Specimen Source	N/A	O	
16	Ordering Provider	N/A	O	Component 1= Physician Code Component 2 = Last Name Component 3 = First Name Example: GRT1^GRANT^JOHN
17	Entering Organization	N/A	NS	

18	Placer Field	N/A	O	
19	Placer Field	N/A	O	
20	N/A	N/A	N/A	
21	N/A	N/A	N/A	
22	N/A	N/A	N/A	
23	N/A	N/A	N/A	
24	N/A	N/A	N/A	
25	Result Status	1	R	

4.1.5 NTE – Notes and Comments

Sequence #	Name	Max Length	Field Requirement	Notes
1	Set ID – NTE	4	O	
2	Source of Comment	8	O	
3	Comment	2000	O	

4.2 Message Acknowledgement

The Message Acknowledgement transaction is separated into the following two segments:

1 MSH – Message Header
2 MSA – Message Acknowledgement

The following table outlines the contents of each segment. Each segment is required in the order transaction.

Legend
R – Required
O – Optional
NS – Not Supported

4.2.1 MSH – Message Header

Sequence #	Name	Max Length	Field Requirement	Notes
1	Field Separator	1	R	|
2	Encoding Characters	4	R	component delimiter (^) repeat delimiter (~) escape character (\) subcomponent delimiter (&)
3	Sending Application	50	R	
4	Sending Facility	50	O	
5	Receiving Applications	50	O	
6	Receiving Facility	50	O	
7	Date/Time of Message	12	R	CCYYMMDDHHMM
8	Security	N/A	NS	
9	Message Type	7	R	Contains message type and trigger event.
10	Message Control ID	15	R	This sequence is valued with a number that is unique to that message. The receiving system will echo this value in the MSA message.
11	Processing ID	3	R	T for Training or P for Production
12	Version ID	8	R	Specifies the HL7 version in use

13	Sequence Number	N/A	NS	
14	Continuation Pointer	N/A	NS	
15	Accept Acknowledgement Type	N/A	NS	
16	Application Acknowledgement Type	N/A	NS	
17	Country Code	N/A	NS	
18	Character Set	N/A	NS	
19	Principal Language of Message	N/A	NS	

4.2.2 MSA – Message Acknowledgement

Sequence #	Name	Max Length	Field Requirement	Notes
1	Acknowledgement Code	2	R	Indicates whether positive (ACK) or negative (NAK) acknowledgment. CA = ACK CR = NAK
2	Message Control ID	15	R	The number sent in MSH; 10 of the inbound message is echoed back to the client in this sequence.
3	Text Message	100	O	This field may be valued but only if returning a NAK and only under certain error conditions. A description of the error may be included here for troubleshooting purposes
4	Expected Sequence Number	N/A	NS	HistoTrac does not use sequence number protocol
5	Delayed ACK Type	N/A	NS	
6	Error Condition	N/A	NS	